KENNESAW STATE UNIVERSITY
COLLEGE OF COMPUTING AND SOFTWARE ENGINEERING
School of Data Science and Analytics

Corn in Context: Linking Meteorology, Fertilizer, and Yield Outcomes

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Presenters Information

Introduction

Background

- Corn accounts for over 95% of animal feed grain and 10% of ethanol across the US
- In 2019-2020, the US exported over 340,000 million metric tons of corn, leading it to be the number 1 corn exporter in the world
- However, several environmental agencies have noted that corn yield has been heavily impacted by drought and heavy precipitation

Purpose

- **To analyze whether environmental challenges and a selected agricultural measure can predict corn yield on an annual basis**

Methods

Data Wrangling:

- Our dataset was combined and gathered from two main sources including United States Department of Agriculture – National Agricultural Statistics Service (USDA-NASS) and National Oceanic and Atmospheric Association – National Center for Environmental Information (NOAA-NCEI)
- Non-disclosed values, notated by (D), were removed
- Data Mapping: 1,423 Counties → 114 Climate Divisions across 16 states including Alabama, Arkansas, Delaware, Florida, Georgia, Kentucky, Louisiana, Maryland, Mississippi, North Carolina, Oklahoma, South Carolina, Tennessee, Texas, Virginia, and West Virginia

Statistical Analysis:

Power Analysis

- A Power Analysis was assessed to determine the number of predictors that could be placed in the model without producing Type II error, based on available sample size

Exploratory Data Analysis

- EDA was performed to assess data quality and variable relationships

Multiple Linear Regression Analysis

- Six multiple linear regression models were assessed. The best model, based on R^2, applied to the analysis
- Model: Yield ~ Corn Fertilizer Expense + Average Cooling Temperature + Palmer Drought Severity Index Category + Precipitation
- Cook's D, DFBetas, DFFits, Studentized Residuals were evaluated to check outliers, leverage, and influential observations
  - Removed Climate Division IDs: 308, 1608, 3801, 4106, 4004, and 803
- Assumption Check : Normality, Linearity, Homoscedasticity
  - Supported after outliers and influential treatment
- Cluster-Robust Variance Estimation (*clubSandwich package*) was applied, using CR2 estimator, to solve for clustering observations
- Multicollinearity was checked by assessing VIF
  - All predictors have VIF < 2
- Partial F-Test was examined to determine if a predictor variable significantly improved the regression model

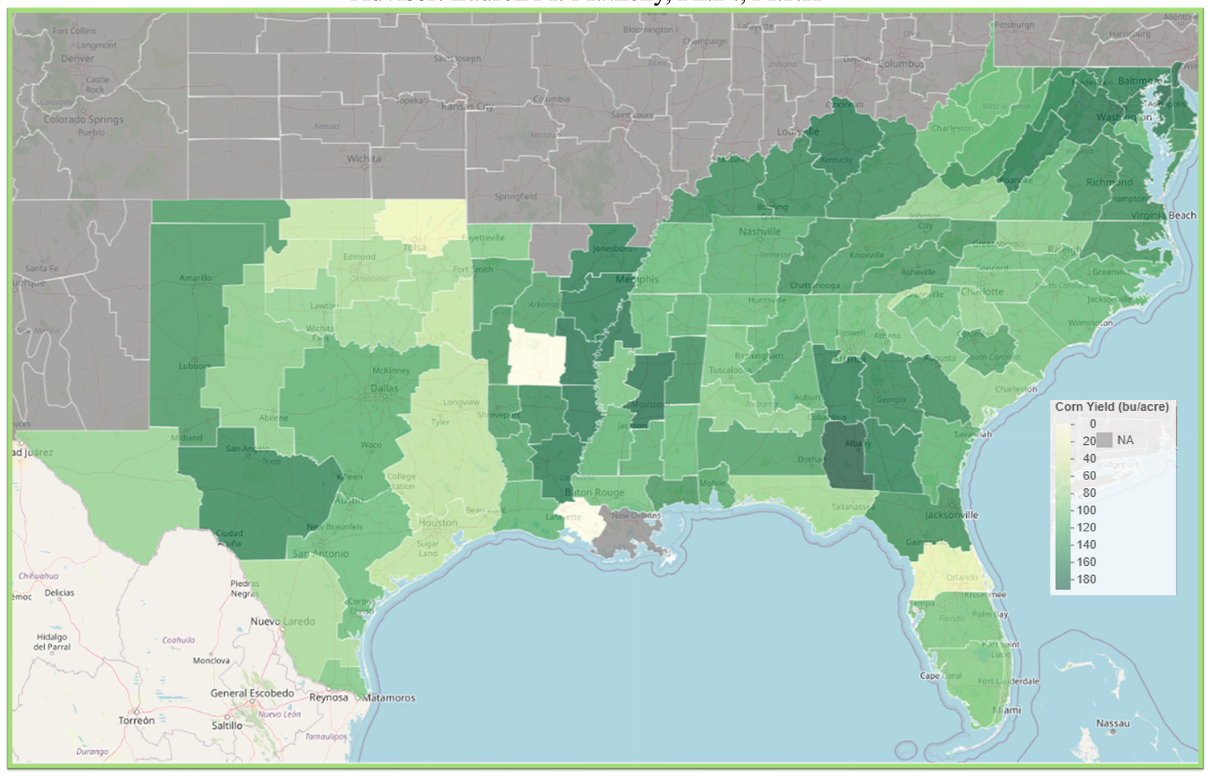


Figure 1: Corn Yield Map per Climate Division Across South US States

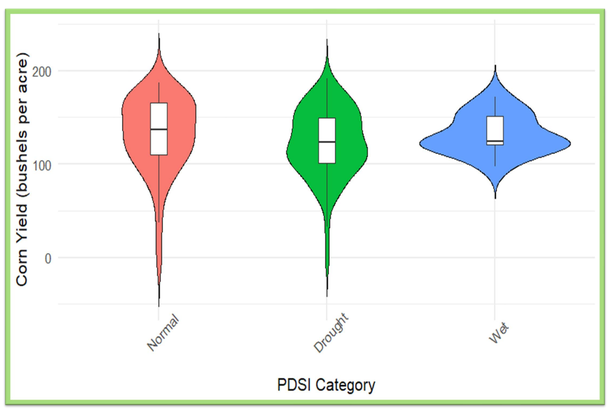


Figure 2: Violin Plot of PDSI Categories

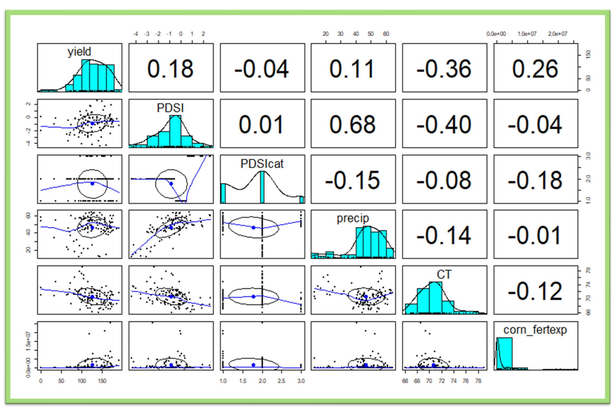


Figure 3: Correlation Matrix Plot of Main Variables

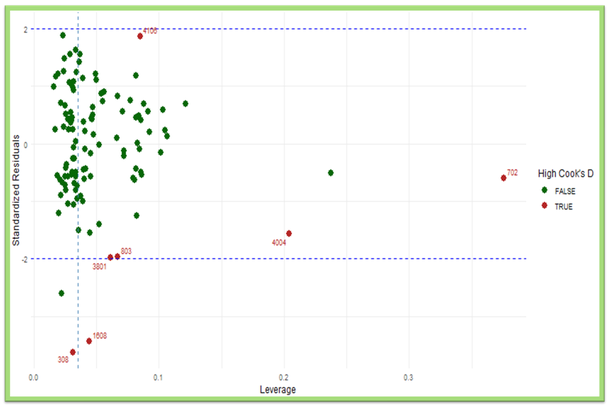


Figure 4: Studentized Residuals vs. Leverage

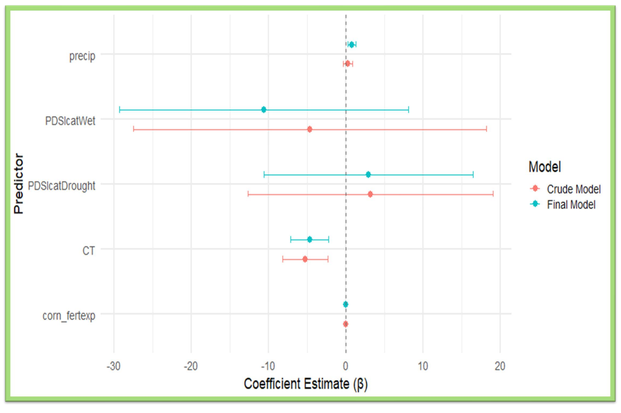


Figure 5: Coefficient Plot of Crude vs. Updated Model

Results

- **Final Model:**
  - Yield = $\beta_0 + \beta_1$(Corn Fertilizer Expense) + β_2(Average Cooling Temperature) + β_3(Precipitation) + β_4(PDSI Category) + ε
  - 23.77% of the variance in Yield is explained by all four predictors, $R^2 = 0.2377$, $f(5, 102) = 7.672$, $p < .001$

Significance Audit — Robust SEs (clubSandwich)

Predictor (term)	Estimate (β)	Robust SE	p-value	Significance	Direction
(Intercept)	**422.59**	78.48	0.003	**	↑ Positive
Fertilizer Expenses	**0.00**	0.00	0.090		ns
Cooling Temperature (F)	**-4.70**	1.10	0.007	**	↓ Negative
PDSI: Dry vs Baseline	**2.93**	7.79	0.714		ns
PDSI: Wet vs Baseline	**-10.60**	11.71	0.401		ns
Precipitation (in)	**0.74**	0.42	0.166		ns

Model R^2 = 0.2733 | Adj. R^2 = 0.2377 | N = 108 • Codes: *** p<0.001, ** p<0.01, * p<0.05

- **Partial F-Test:**
  - Precipitation: 7.3%
  - Fertilizer Expense: 9.5%
  - PDSI Categories: 1.04%
  - Cooling Temperature: 13.19%

Conclusion

- The implemented clubSandwich CR2 estimator provides reasonable bias-reduced cluster-robust variance estimator suited for datasets with small number of clusters
- In the future, it may be best to consider other crops instead of focusing on a singular one
- This model could be utilized regionally for crops produced in a specific geographical area
- This analysis may be more beneficial if it could be run as a time series model to see whether the specific growing season is affected by these variables rather than the entire year
- **Overall, both precipitation and cooling temperature can be used to predict ~23% of corn yield on an annual basis**

Limitations

- **Data Format/Availability:** During the data collection process, most of the available climate datasets were in Geographic Information System (GIS) or graphical formats, or the datasets available had limited information by county level. Additionally, many of these climate variables involved temporal components that are better suited for time-series or longitudinal analyses.
- **Statistical Methods:** Our current model contains clusters that would be more appropriately analyzed using a multimodal regression approach. However, we have not yet had the opportunity to learn or apply this method.